

Using Encryption to Enhance Confidentiality and Integrity (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 05

Student:

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Email:

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Time on Task:

7 hours, 13 minutes

Progress:

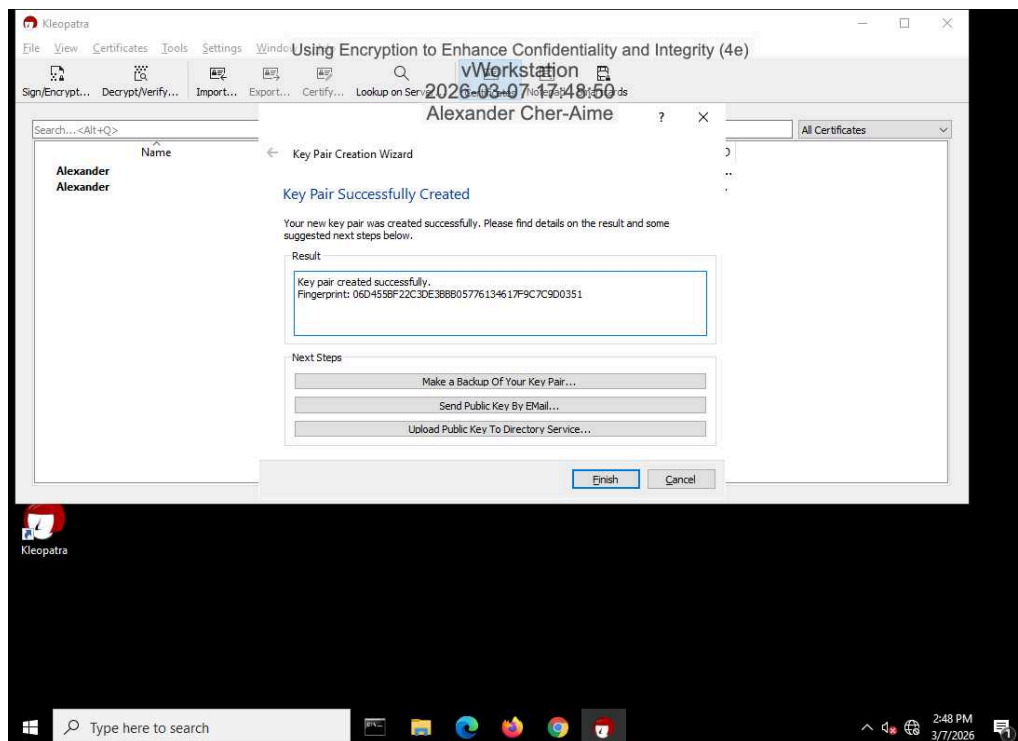
100%

Report Generated: Sunday, March 8, 2026 at 4:51 AM

Section 1: Hands-On Demonstration

Part 1: Create and Exchange Asymmetric Encryption Keys

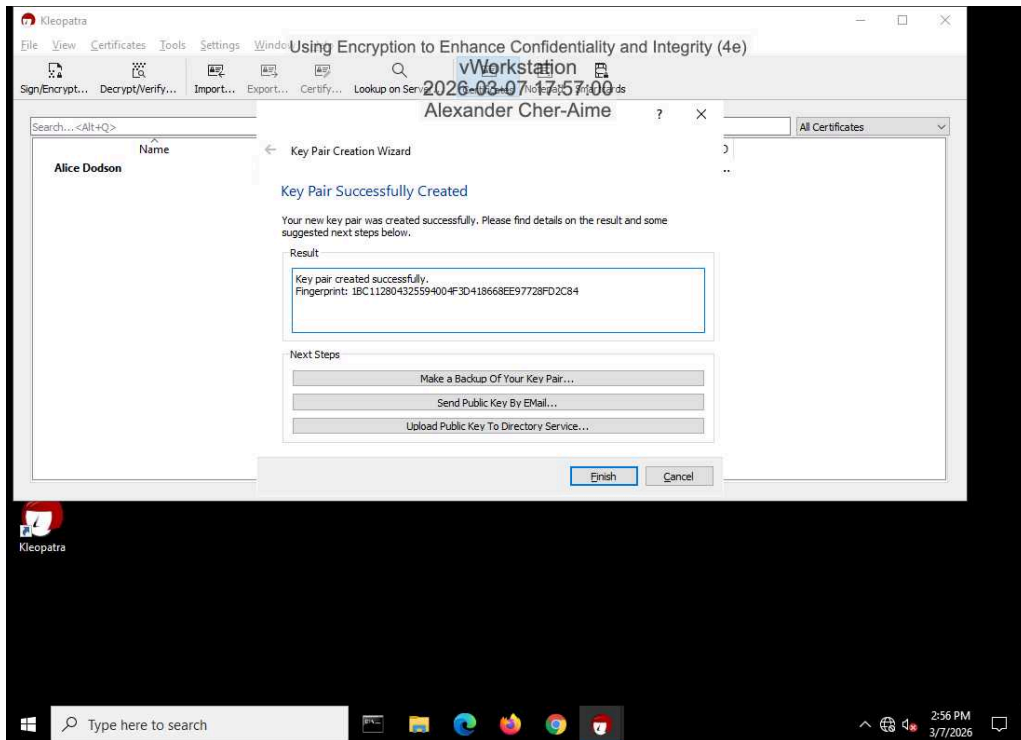
9. Make a screen capture showing the **fingerprint** for your key pair.



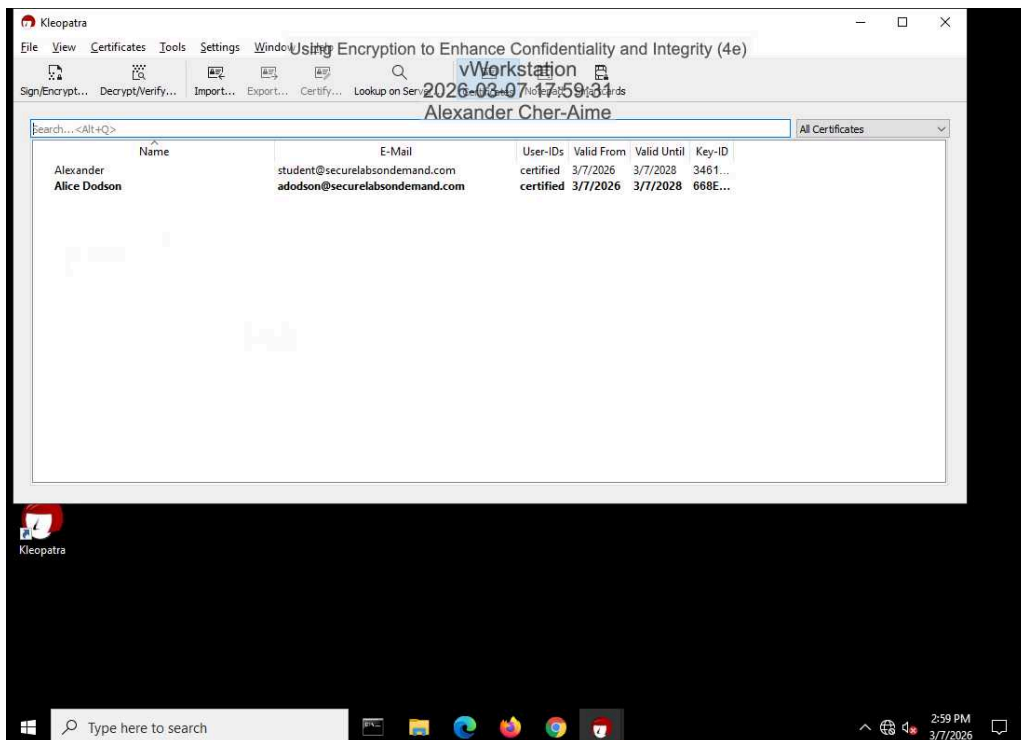
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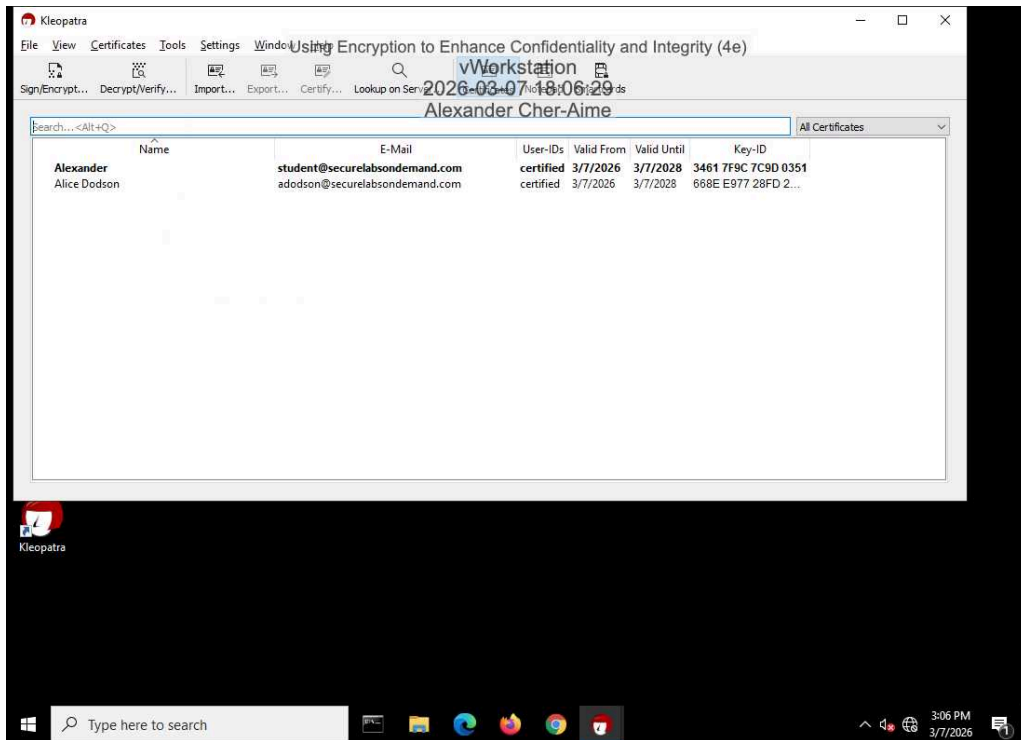
22. Make a screen capture showing the **fingerprint** for Alice's key pair.



30. Make a screen capture showing your public key in Alice's certificate cache.



35. Make a screen capture showing Alice's public key in your certificate cache.

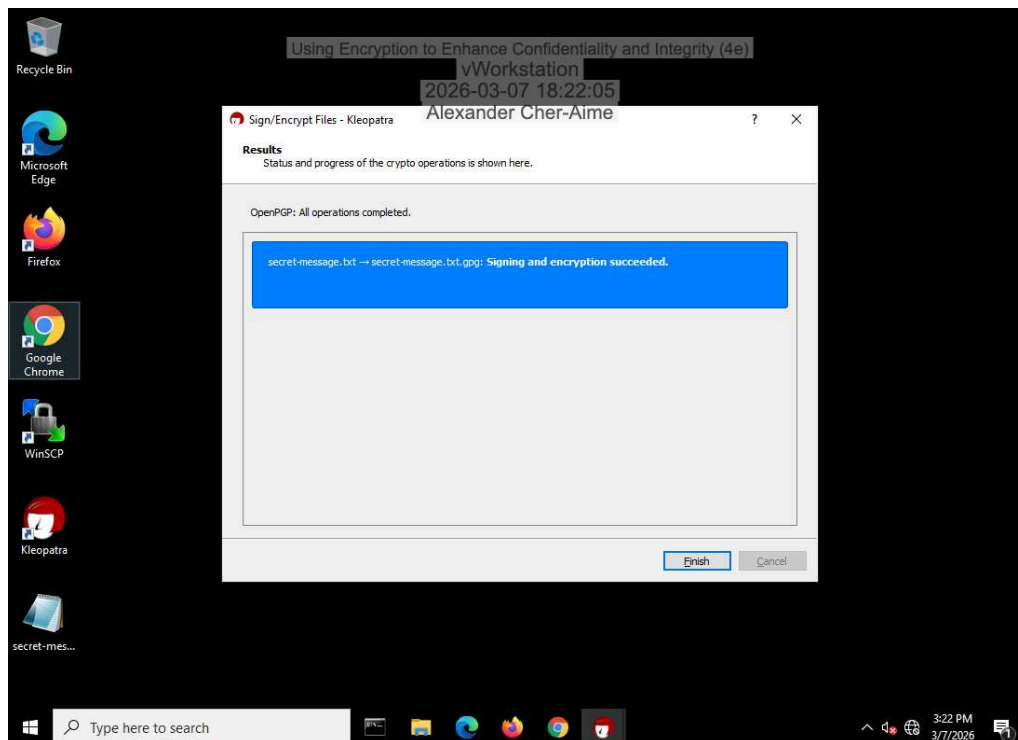


Part 2: Encrypt a File Using Asymmetric Encryption

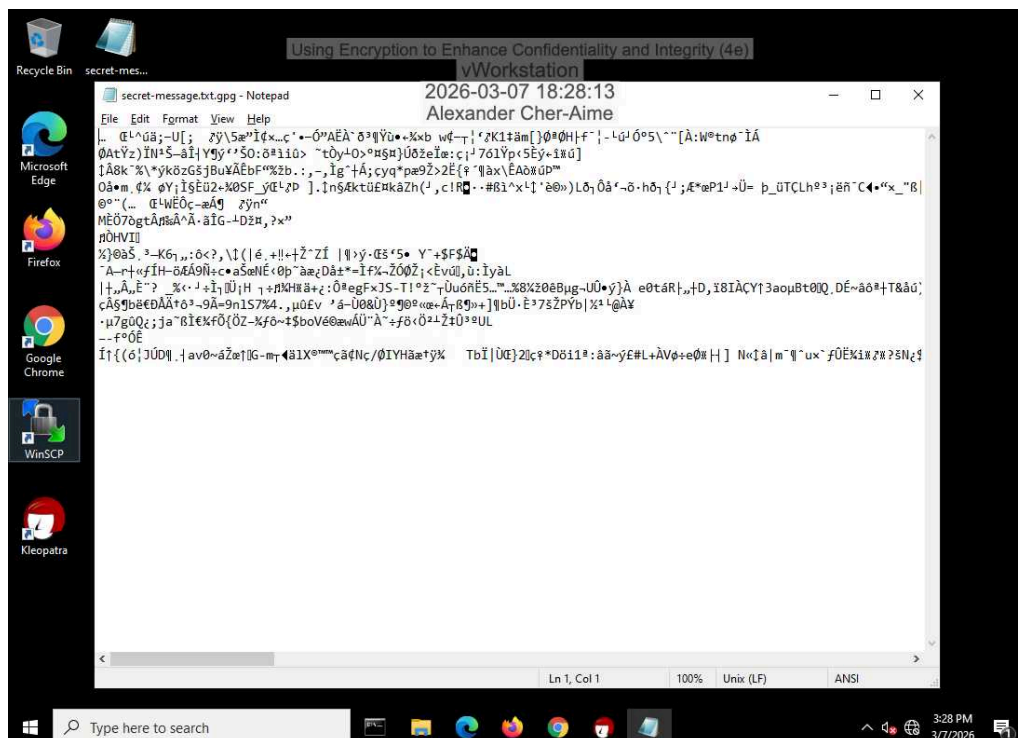
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9. Make a screen capture showing the **successful signing and encryption message**.

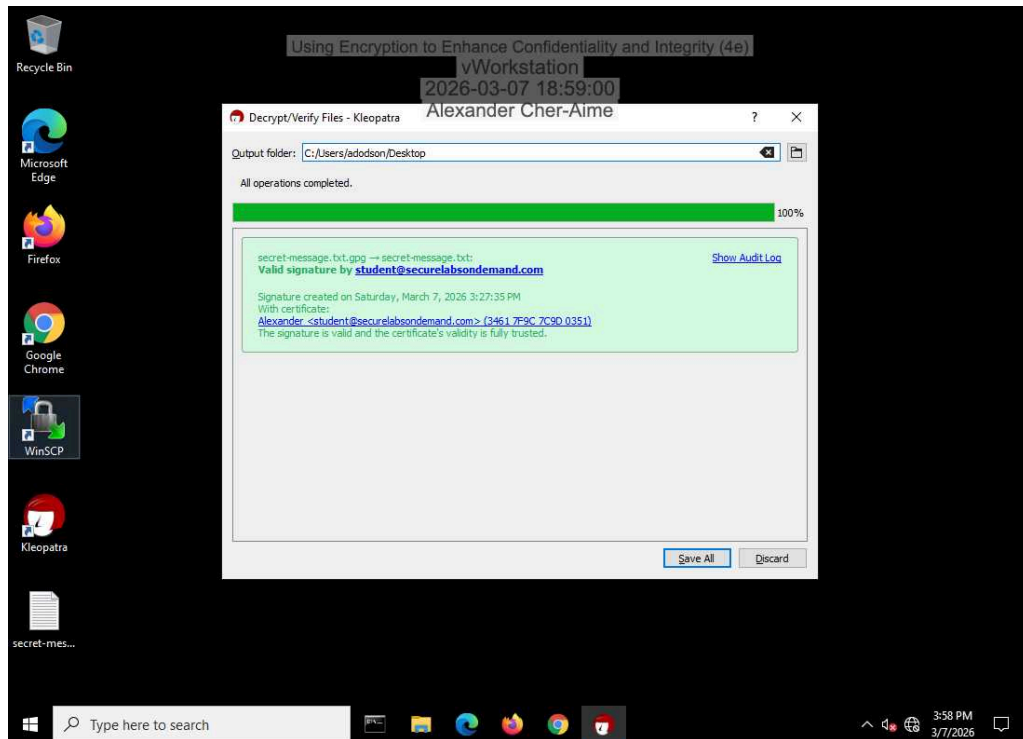


12. Make a screen capture showing the **ciphertext**.



Part 3: Decrypt a File Using Asymmetric Encryption

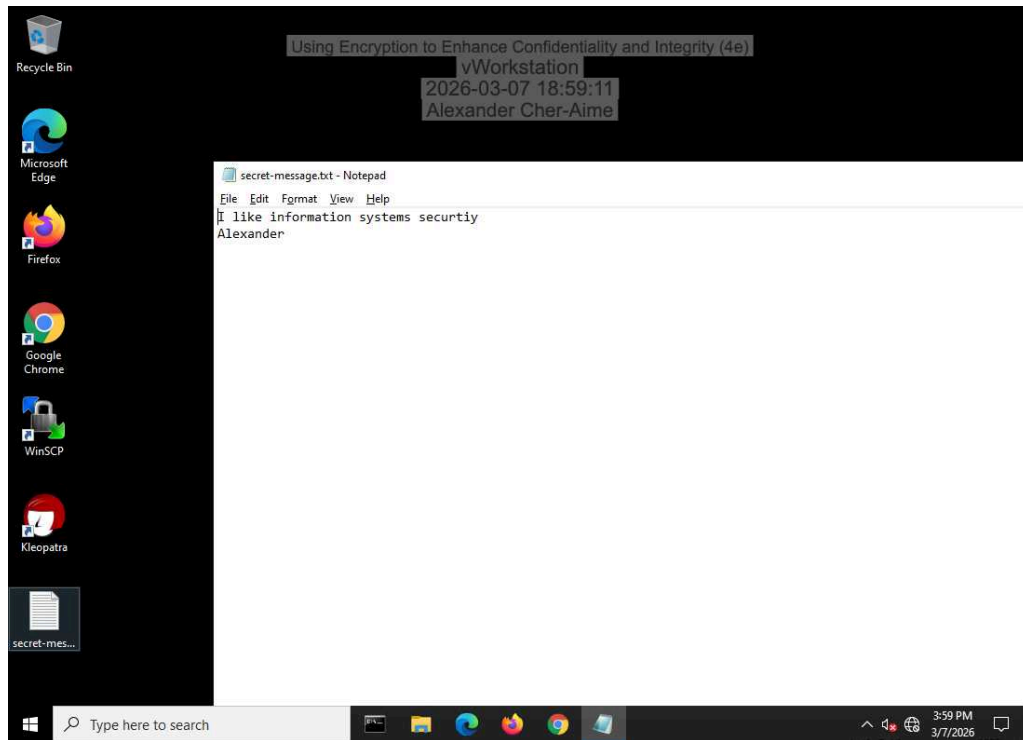
15. Make a screen capture showing the **Decrypt/Verify Files** window.



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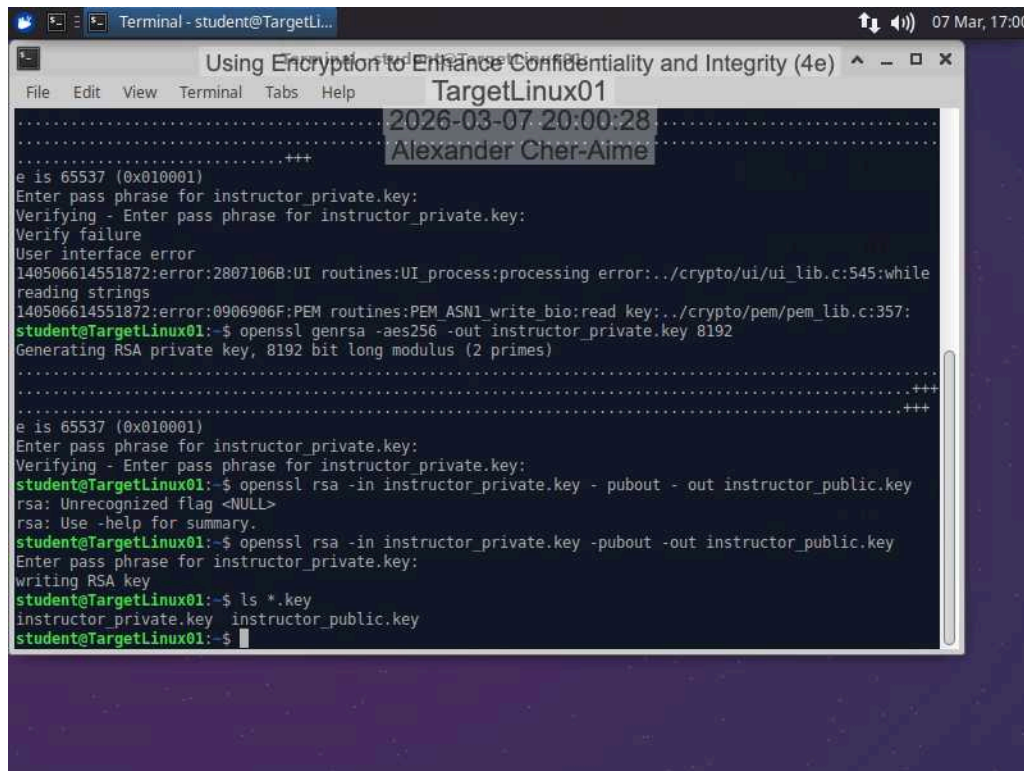
18. Make a screen capture showing the **decrypted secret-message.txt** file in Notepad.



Section 2: Applied Learning

Part 1: Create an Asymmetric Key Pair

10. Make a screen capture showing the instructor's key pair files.



The screenshot shows a terminal window titled "Terminal - student@TargetLinux01" with a window manager overlay. The overlay includes the title "Using Encryption to Enhance Confidentiality and Integrity (4e)", the window name "TargetLinux01", a timestamp "2026-03-07 20:00:28", and the user name "Alexander Cher-Aime". The terminal output shows the user attempting to generate a private key with the command `openssl genrsa -aes256 -out instructor_private.key 8192`. This command results in several error messages related to the UI routines. The user then successfully generates the private key with `openssl genrsa -out instructor_private.key 8192`. Finally, the user generates the public key with `openssl rsa -in instructor_private.key -pubout -out instructor_public.key`. The terminal ends with a `ls *.key` command showing the files `instructor_private.key` and `instructor_public.key`.

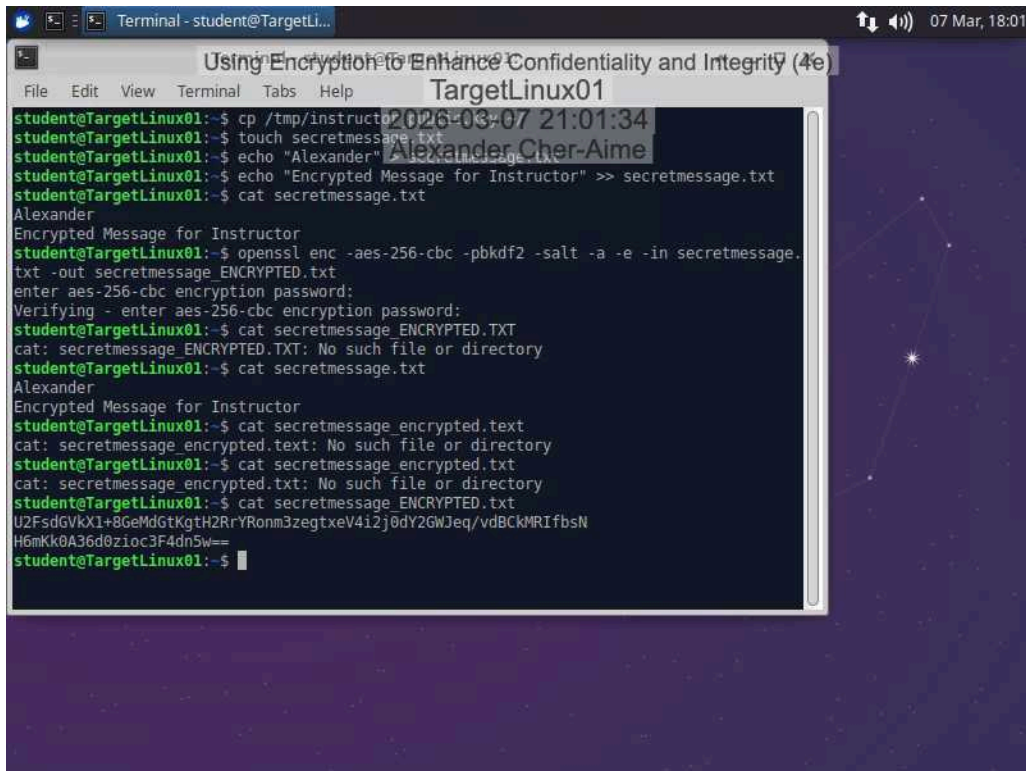
```
.....++
e is 65537 (0x010001)
Enter pass phrase for instructor_private.key:
Verifying - Enter pass phrase for instructor_private.key:
Verify failure
User interface error
140506614551872:error:28071068:UI routines:UI_process:processing error:../crypto/ui/ui_lib.c:545:while
reading strings
140506614551872:error:0906906F:PEM routines:PEM ASN1 write_bio:read key:../crypto/pem/pem_lib.c:357:
student@TargetLinux01:~$ openssl genrsa -aes256 -out instructor_private.key 8192
Generating RSA private key, 8192 bit long modulus (2 primes)
.....++
.....++
e is 65537 (0x010001)
Enter pass phrase for instructor_private.key:
Verifying - Enter pass phrase for instructor_private.key:
student@TargetLinux01:~$ openssl rsa -in instructor_private.key -pubout -out instructor_public.key
rsa: Unrecognized flag <NULL>
rsa: Use -help for summary.
student@TargetLinux01:~$ openssl rsa -in instructor_private.key -pubout -out instructor_public.key
Enter pass phrase for instructor_private.key:
writing RSA key
student@TargetLinux01:~$ ls *.key
instructor_private.key  instructor_public.key
student@TargetLinux01:~$
```

Part 2: Encrypt a File Using Symmetric Encryption

11. Document the password you used to symmetrically encrypt the file.

alexander

13. Make a screen capture showing the ciphertext in the `secretmessage_ENCRYPTED.txt` file.

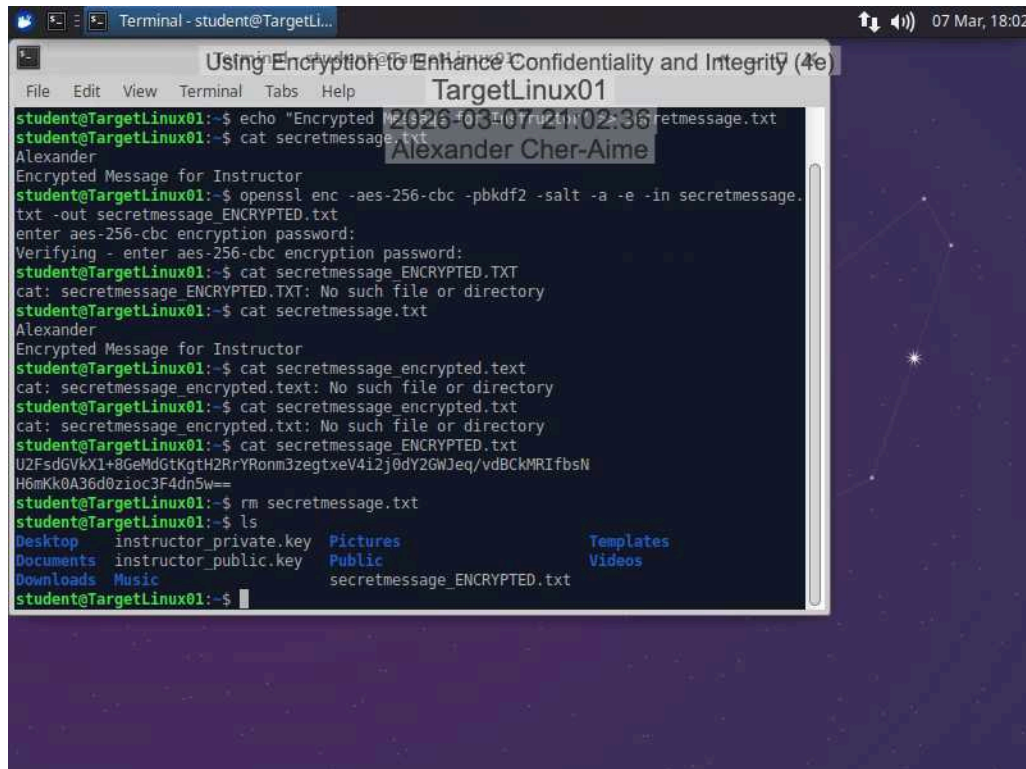


```
student@TargetLinux01:~$ cp /tmp/instructor/secretmessage.txt .
student@TargetLinux01:~$ touch secretmessage.txt
student@TargetLinux01:~$ echo "Alexander" >> secretmessage.txt
student@TargetLinux01:~$ echo "Encrypted Message for Instructor" >> secretmessage.txt
student@TargetLinux01:~$ cat secretmessage.txt
Alexander
Encrypted Message for Instructor
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.
txt -out secretmessage_ENCRYPTED.txt
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.TXT
cat: secretmessage_ENCRYPTED.TXT: No such file or directory
student@TargetLinux01:~$ cat secretmessage.txt
Alexander
Encrypted Message for Instructor
student@TargetLinux01:~$ cat secretmessage_encrypted.txt
cat: secretmessage_encrypted.txt: No such file or directory
student@TargetLinux01:~$ cat secretmessage_encrypted.txt
cat: secretmessage_encrypted.txt: No such file or directory
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.txt
U2FsdGVkXl+8GeMdGtKgtH2RrYRonm3zegtXeV4i2j0dY2GWJeq/vdBCKMRIbfsN
H6mKk0A36d0zioc3F4dn5w==
student@TargetLinux01:~$
```


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16. Make a screen capture showing the output of the ls command.



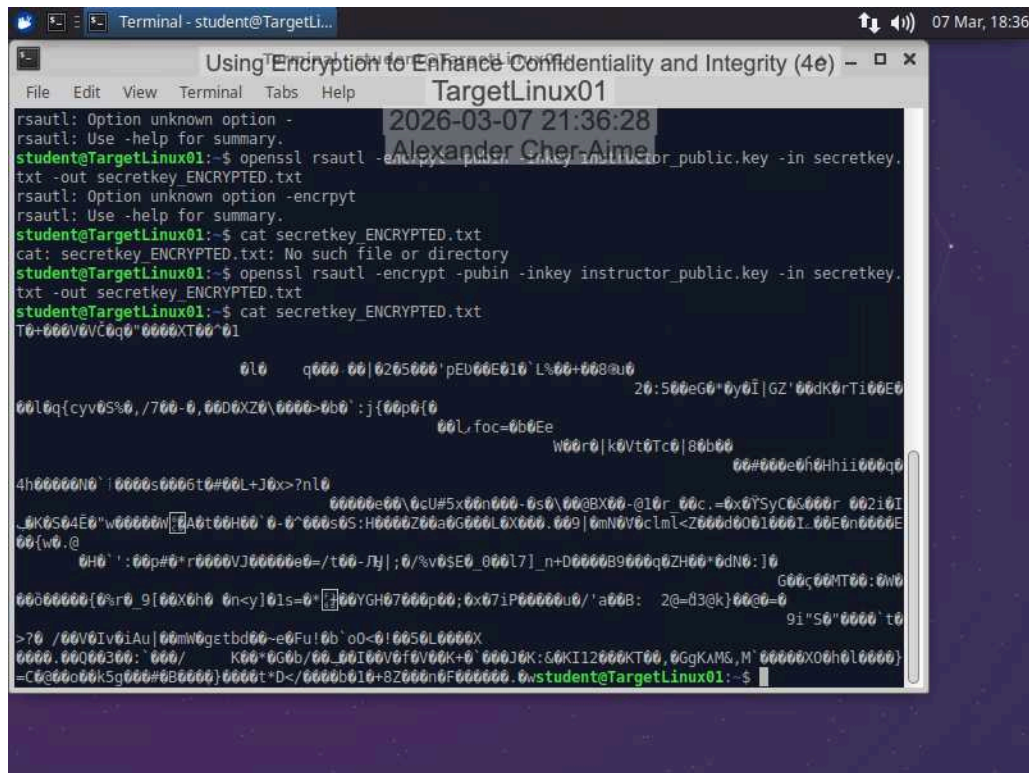
```
student@TargetLinux01:~$ echo "Encrypted Message for Instructor"
Encrypted Message for Instructor
student@TargetLinux01:~$ cat secretmessage.txt
Alexander
Encrypted Message for Instructor
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.
txt -out secretmessage_ENCRYPTED.TXT
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.TXT
cat: secretmessage_ENCRYPTED.TXT: No such file or directory
student@TargetLinux01:~$ cat secretmessage.txt
Alexander
Encrypted Message for Instructor
student@TargetLinux01:~$ cat secretmessage_encrypted.txt
cat: secretmessage_encrypted.txt: No such file or directory
student@TargetLinux01:~$ cat secretmessage_encrypted.txt
cat: secretmessage_encrypted.txt: No such file or directory
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.TXT
U2FsdGVkX1+8GeMdGtKgtH2RrYRonm3zegtxeV4i2j0dY2GWJeq/vdBckMRIfbsN
H6mKk0A36d0zioc3F4dn5w==
student@TargetLinux01:~$ rm secretmessage.txt
student@TargetLinux01:~$ ls
Desktop    instructor_private.key  Pictures          Templates
Documents  instructor_public.key   Public            Videos
Downloads  Music                  secretmessage_ENCRYPTED.TXT
```

Part 3: Transfer and Decrypt a File Using Hybrid Cryptography

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6. Make a screen capture showing the encrypted contents of the `secretkey_ENCRYPTED.txt` file.



The screenshot shows a terminal window titled "Terminal - student@TargetLinux01" with a timestamp of "07 Mar, 18:36". The window displays the following commands and output:

```
rsautl: Option unknown option -
rsautl: Use -help for summary.
student@TargetLinux01:~$ openssl rsautl -encrypt -pubin -inkey instructor_public.key -in secretkey.
txt -out secretkey_ENCRYPTED.txt
rsautl: Option unknown option -encrypt
rsautl: Use -help for summary.
student@TargetLinux01:~$ cat secretkey_ENCRYPTED.txt
cat: secretkey_ENCRYPTED.txt: No such file or directory
student@TargetLinux01:~$ openssl rsautl -encrypt -pubin -inkey instructor_public.key -in secretkey.
txt -out secretkey_ENCRYPTED.txt
student@TargetLinux01:~$ cat secretkey_ENCRYPTED.txt
T0+000V0VC0q0"0000XT00^01

010 q000 00|0205000'pE000E010'L%00+0080u0
00L0q{cyv0S%,/700-0,0000XZ0\0000>0b0':j{00p0f0 20:500eG0"0y0I|GZ'00dK0rTi00E0
00L,foc=0b0Ee W00r0|k0vt0Tc0|80b00
00#000e0f0Hhii000q0
4h00000N0'0000s0006t0#00L+J0x>?nL0
00000e00\0cU#5x00n000-0s0\000BX00-@10r 00c.=0x0YSyC0G000r 002i0I
_8K0S04E0"w00000n[A0t00H00'0-0^000s0S:H0000Z00a0G000L0X000.009|0mN0V0cLmL<Z00d00010001.00E0n0000E
00{w0.@
0H0':00p#0*r0000VJ00000000=/t00-7y|;0/%v0SE0_000L7]_n+00000B9000q0ZH00"0dN0:]0
G00c00MT00:0w0
00000000{0sr0_9[00X0h0 0n<y]01s=0"00YGH07000p00;0x071P00000u0/'a00B: 2@=03@k}0000=0
9i"S0"0000't0
>70 /00V0Iv0iAu|00mW0getbd00-e0Fu!0b'oD<0!0050L0000X
0000.00Q00300:'000/ K00"0G0b/00_00I00V0f0V00K+0'000J0K:60KI12000KT00,0GgKAMS,M'00000X00h0L0000}
=C0@00000k5g000#0B0000}0000t*D</0000b010+8Z000n0F000000.0wstudent@TargetLinux01:~$
```

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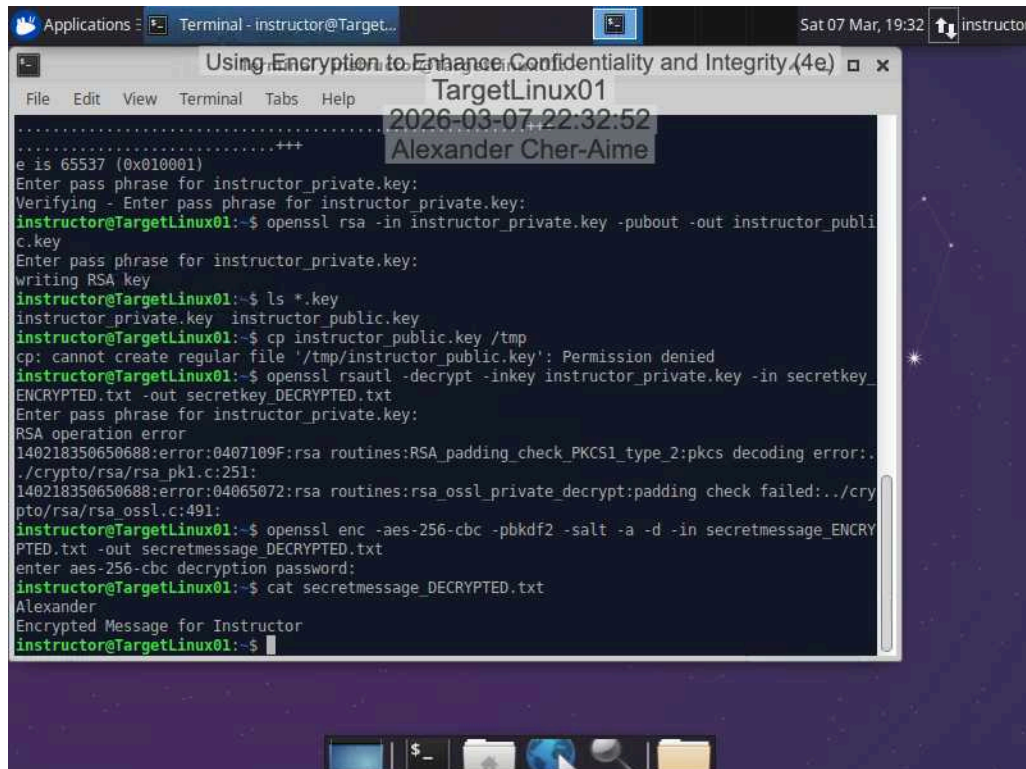
17. Make a screen capture showing the **decrypted contents of the secretkey_DECRYPTED.txt file.**

```
cat: instructor_private.key: No such file or directory
instructor@TargetLinux01:~$ openssl genrsa -aes256 -out instructor_private.key 8192
Invalid command 'genrsa'; type 'help' for a list.
instructor@TargetLinux01:~$ openssl genrsa -aes256 -out instructor_private.key 8192
Generating RSA private key, 8192 bit long modulus (2 primes)
.....+++
.....+++
e is 65537 (0x010001)
Enter pass phrase for instructor_private.key:
Verifying - Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ openssl rsa -in instructor_private.key -pubout -out instructor_public.key
Enter pass phrase for instructor_private.key:
writing RSA key
instructor@TargetLinux01:~$ ls *.key
instructor_private.key  instructor_public.key
instructor@TargetLinux01:~$ cp instructor_public.key /tmp
cp: cannot create regular file '/tmp/instructor_public.key': Permission denied
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.key -in secretkey_ENCRYPTED.txt -out secretkey_DECRYPTED.txt
Enter pass phrase for instructor_private.key:
RSA operation error
140218350650688:error:0407109F:rsa routines:RSA_padding_check_PKCS1_type_2:pkcs decoding error:../crypto/rsa/rsa_pk1.c:251:
140218350650688:error:04065072:rsa routines:rsa_ossl_private_decrypt:padding check failed:../crypto/rsa/rsa_ossl.c:491:
instructor@TargetLinux01:~$
```

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21. Make a screen capture showing the contents of the `secretmessage_DECRYPTED` file.



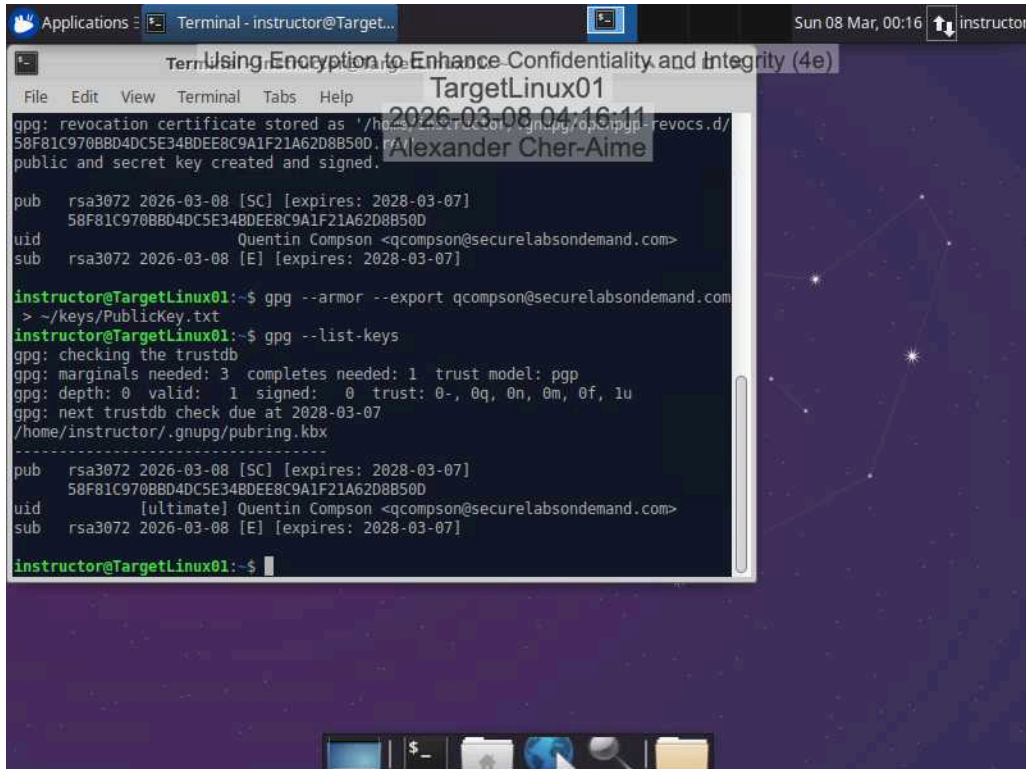
The screenshot shows a terminal window titled "Using Encryption to Enhance Confidentiality and Integrity (4e)" with a subtitle "TargetLinux01". The terminal output shows the following commands and results:

```
.....+++
e is 65537 (0x010001)
Enter pass phrase for instructor_private.key:
Verifying - Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ openssl rsa -in instructor_private.key -pubout -out instructor_public.key
Enter pass phrase for instructor_private.key:
writing RSA key
instructor@TargetLinux01:~$ ls *.key
instructor_private.key  instructor_public.key
instructor@TargetLinux01:~$ cp instructor_public.key /tmp
cp: cannot create regular file '/tmp/instructor_public.key': Permission denied
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.key -in secretkey_ENCRYPTED.txt -out secretkey_DECRYPTED.txt
Enter pass phrase for instructor_private.key:
RSA operation error
140218350650688:error:0407109F:rsa routines:RSA_padding_check_PKCS1_type_2:pkcs decoding error:../crypto/rsa/rsa_pk1.c:251:
140218350650688:error:04065072:rsa routines:rsa_ossl_private_decrypt:padding check failed:../crypto/rsa/rsa_ossl.c:491:
instructor@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -d -in secretmessage_ENCRYPTED.txt -out secretmessage_DECRYPTED.txt
enter aes-256-cbc decryption password:
instructor@TargetLinux01:~$ cat secretmessage_DECRYPTED.txt
Alexander
Encrypted Message for Instructor
instructor@TargetLinux01:~$
```

Section 3: Challenge and Analysis

Part 1: Digitally Sign a Document Using GPG

Make a screen capture showing the **key fingerprint** for the key pair you generated in this part of the lab.



```
gpg: revocation certificate stored as '/home/instructor/.gnupg/openpgp-revocs.d/58F81C970BBD4DC5E348DEE8C9A1F21A62D8B50D.rsa3072.asc'
public and secret key created and signed.

pub  rsa3072 2026-03-08 [SC] [expires: 2028-03-07]
     58F81C970BBD4DC5E348DEE8C9A1F21A62D8B50D
uid          Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2026-03-08 [E] [expires: 2028-03-07]

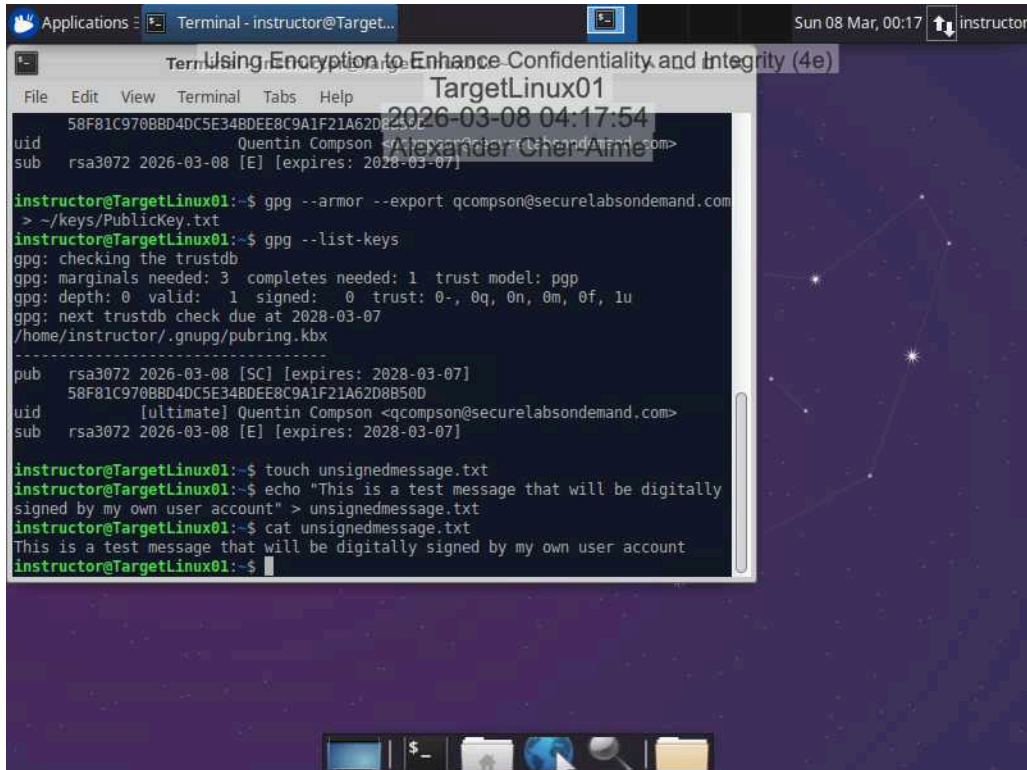
instructor@TargetLinux01:~$ gpg --armor --export qcompson@securelabsondemand.com > ~/keys/PublicKey.txt
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-03-07
/home/instructor/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-03-08 [SC] [expires: 2028-03-07]
     58F81C970BBD4DC5E348DEE8C9A1F21A62D8B50D
uid          [ultimate] Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2026-03-08 [E] [expires: 2028-03-07]

instructor@TargetLinux01:~$
```


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Make a screen capture showing the contents of the unsignedmessage.txt file.



The screenshot shows a terminal window titled "Terminal - instructor@TargetLinux01" with a dark background and light text. The terminal output shows the following commands and results:

```
instructor@TargetLinux01:~$ gpg --armor --export qcompson@securelabsondemand.com > ~/keys/PublicKey.txt
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-03-07
/home/instructor/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-03-08 [SC] [expires: 2028-03-07]
       58F81C9708BD4DC5E34BDEE8C9A1F21A62D8B50D
uid       [ultimate] Quentin Compson <qcompson@securelabsondemand.com>
sub   rsa3072 2026-03-08 [E] [expires: 2028-03-07]

instructor@TargetLinux01:~$ touch unsignedmessage.txt
instructor@TargetLinux01:~$ echo "This is a test message that will be digitally signed by my own user account" > unsignedmessage.txt
instructor@TargetLinux01:~$ cat unsignedmessage.txt
This is a test message that will be digitally signed by my own user account
instructor@TargetLinux01:~$
```

The terminal window is part of a desktop environment with a taskbar at the bottom showing icons for a terminal, file manager, and other applications. The system clock in the top right corner indicates "Sun 08 Mar, 00:17".

Part 2: Verify the Digital Signature Using Kleopatra

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Make a screen capture showing the successful signature verification on the signed message file.

